Factory Manager — Desktop Application

SYNOPSIS

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1. Introduction

Factory Manager is a desktop application developed using Python and the CustomTkinter GUI framework. It is designed to help small-scale manufacturing businesses — particularly wooden box manufacturers — manage their day-to-day operations efficiently without requiring any internet connection or technical knowledge.

The application provides a complete solution for tracking inventory, recording sales, generating invoices, monitoring delivery status, and producing reports. All data is stored locally in an SQLite database, ensuring that information is always available even without internet access.

The core objective of this project is to replace manual record-keeping with a digital system that is simple, fast, and reliable. The application can be packaged as a standalone executable (.exe) file, allowing it to be used on any Windows computer without installing Python or any other software.

2. Objectives

- To develop a desktop application for factory management using Python.
- To eliminate manual record-keeping by providing a digital inventory and sales system.
- To generate professional PDF invoices automatically.
- To track delivery status of customer orders in real time.
- To produce daily and monthly reports exportable to Excel.
- To package the application as a standalone .exe file for easy distribution.
- To provide both Dark and Light mode for better user experience.

3. Modules

Module	Description
Dashboard	Displays real-time overview including total stock items, low stock alerts, total revenue, pending deliveries, and recent activity.

Inventory	Allows adding, editing, and deleting raw materials. Tracks stock levels and sends low-stock warnings. Records purchases from suppliers and automatically updates stock quantities.
Sales	Records customer sales with box type, quantity, price per unit, and auto-calculated total. Supports search and delete functionality.
Billing	Generates professional tax invoices with auto-incremented invoice numbers (INV-0001 format). Invoices can be downloaded as PDF files.
Ready to Deliver	Tracks delivery status (Ready, Dispatched, Delivered) for each customer order. Status can be updated directly from the list.
Reports	Generates daily and monthly reports combining stock, sales, and delivery data. Reports can be exported to Excel (.xlsx) format.

4. Technology Used

Technology	Details
Language	Python 3.x
GUI Framework	CustomTkinter
Database	SQLite3 (built-in)
PDF Generation	fpdf2
Excel Export	pandas + openpyxl
Packaging	PyInstaller (.exe)
IDE	Visual Studio Code
Platform	Windows

5. Conclusion

The Factory Manager application successfully achieves its goal of providing a simple, efficient, and reliable desktop solution for small manufacturing businesses. The project demonstrates the practical use of Python for building real-world business applications.

The application covers all key aspects of factory management including inventory tracking, sales recording, invoice generation, delivery monitoring, and report generation. By packaging it as a standalone executable, it can be deployed on any Windows computer without requiring any technical setup.

Future enhancements could include multi-user network support, barcode scanning integration, and cloud database synchronization for businesses with multiple workstations.